ML peak finding for VFE-board QC traces

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WHAT THIS IS
  Peak finder for VFE-board QC voltage traces, trained only on synthetic toys.
  No real board data was used: the generator encodes the verbally described shape.
  Every number below is measured on held-out TOYS, not on real traces.

WAVEFORM MODEL ENCODED IN THE GENERATOR
  - a train of many small spikes; the last one is P0
  - P1..P4 rise monotonically in amplitude
  - each main peak decays only partway, so the inter-peak floor keeps climbing
  - small irregular structure sits on those floors and must not be reported
  - spiky, asymmetric peaks: Gaussian rise into an exponential decay
  - Gaussian readout noise plus occasional single-sample glitches

METHOD
  Input   one normalised trace, 1024 samples (median-subtracted, max-abs scaled)
  Model   fully convolutional 1D U-Net, 3 down / 3 up levels, kernel 9
  Params  461,717
  Output  5 per-sample heatmap channels, one per peak; position = argmax, confidence = max sigmoid
  Loss    BCEWithLogits against Gaussian targets, sigma = 4.0 samples
  Data    8000 training toys, 1000 held-out toys, 25 epochs, device cuda
  Runtime 125.3 s total

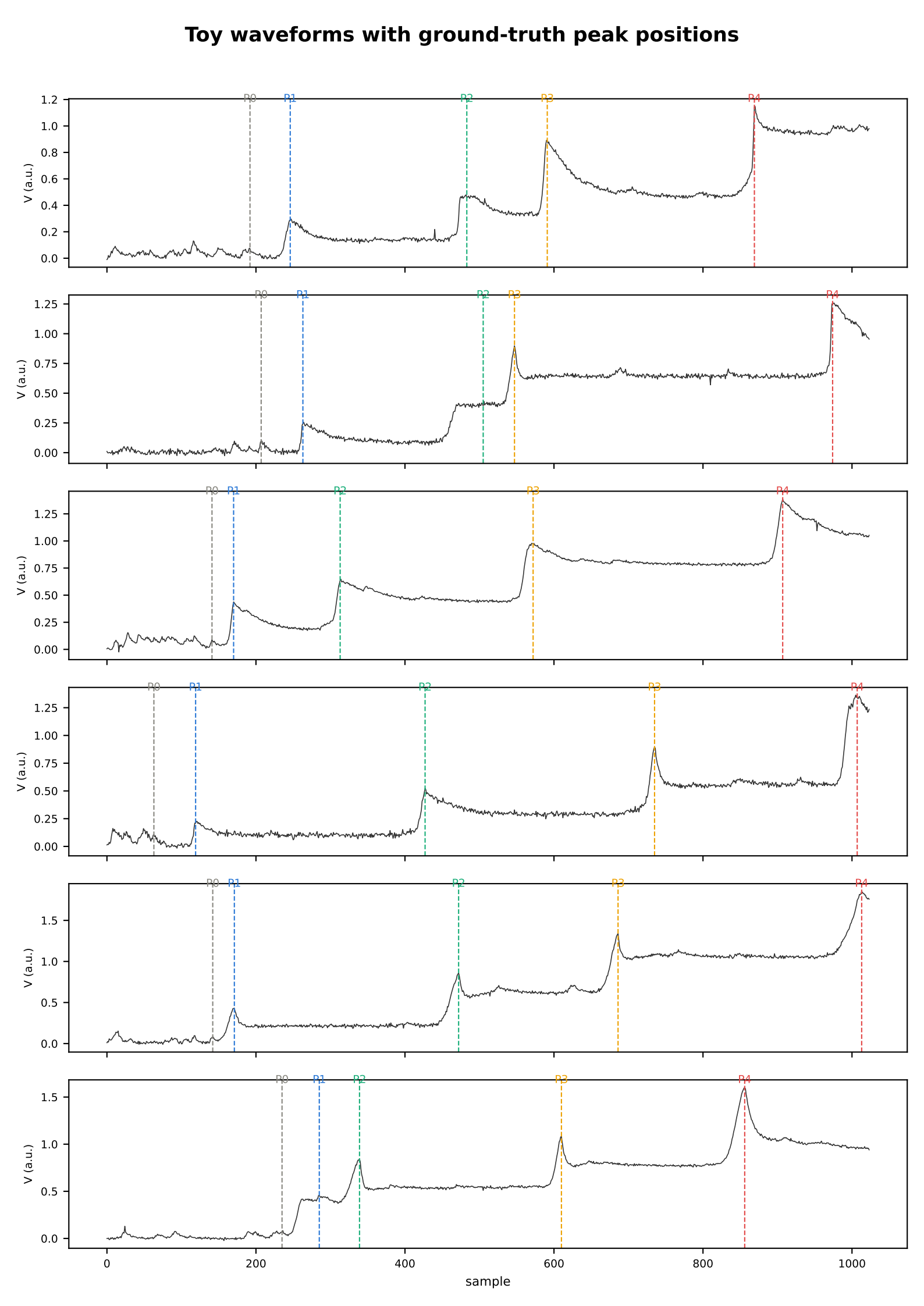
RESULTS ON HELD-OUT TOYS
  all five peaks within 2 samples : 76.8 % of traces
  any peak off by > 15 samples    : 2.00 % of traces

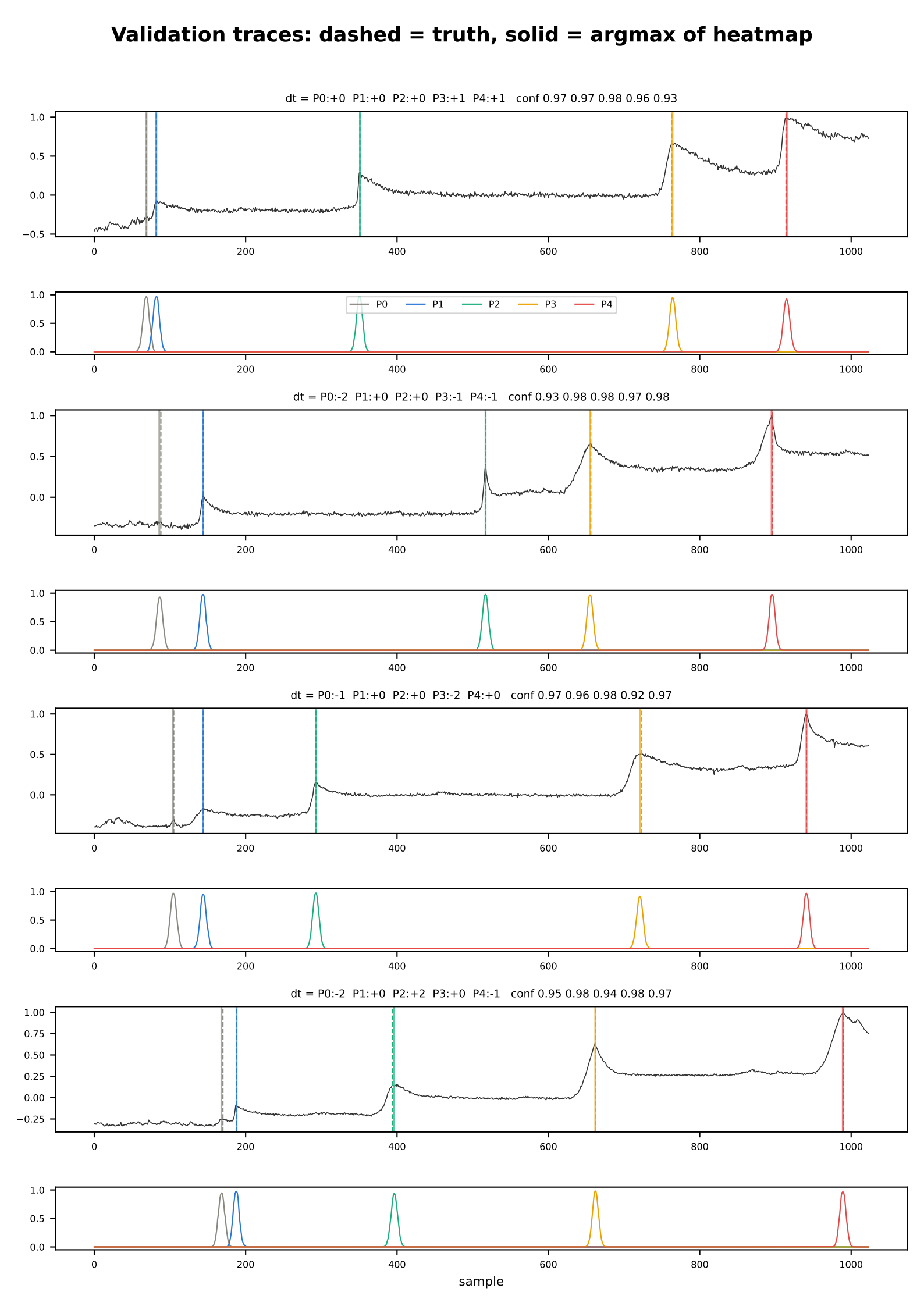
  peak   median|dt|   mean|dt|   p95|dt|   within 2   gross miss   mean conf
  P0          1.00       0.94       2.0      96.0%        0.80%      0.952
  P1          0.00       0.96       4.0      90.9%        0.30%      0.939
  P2          0.00       1.52       3.0      94.6%        0.40%      0.946
  P3          0.00       1.13       2.0      95.9%        0.20%      0.952
  P4          0.00       1.00       2.0      95.5%        0.40%      0.954

LIMITS
  - Trained and tested on toys only; real-board performance is UNVERIFIED.
  - The generator's amplitudes, noise level and peak widths are placeholders,
    chosen by hand, not fitted to measured traces.
  - The model assumes exactly five peaks per trace and always returns five.
    A board missing a peak will still get five positions, flagged only by low confidence.

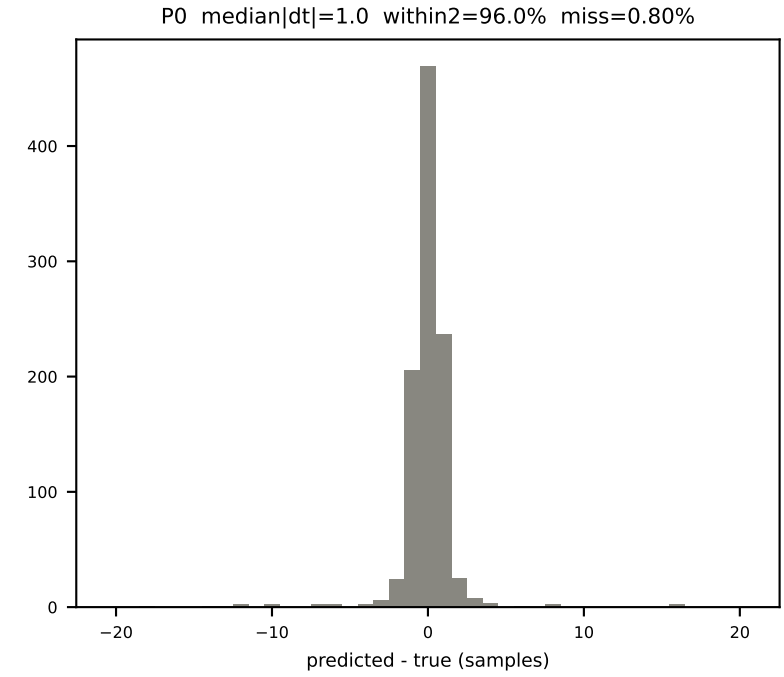
NEXT STEP WITH REAL DATA
  1. Fit the generator config to measured traces (amplitudes, noise, widths, floor steps).
  2. Hand-label a few hundred real traces, especially the ones the current finder gets wrong.
  3. Fine-tune on real labels and report the same table on real held-out traces.

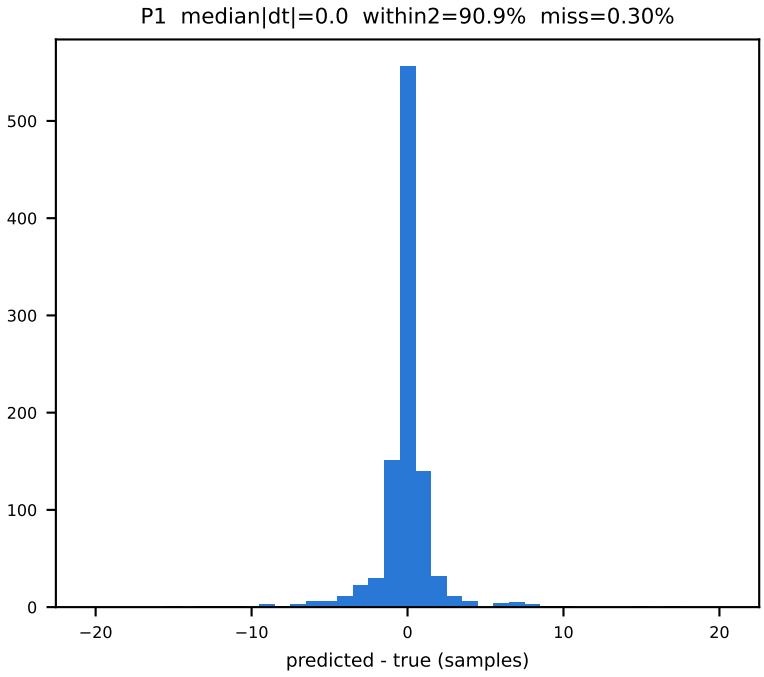
GENERATOR CONFIG USED
  n_samples          1024
  peak_window        (0.14, 0.9)
  p0_gap             (10.0, 60.0)
  n_pre_spikes       (8, 20)
  pre_amp            (0.01, 0.06)
  p0_amp             (0.04, 0.09)
  main_amp           (0.25, 1.0)
  rise_width         (1.5, 14.0)
  decay_tau          (2.5, 30.0)
  shape_bias         (0.5, 2.0)
  floor_frac         (0.25, 0.55)
  step_width         (3.0, 10.0)
  n_decoys           (2, 6)
  decoy_frac         (0.04, 0.18)
  noise_sigma        (0.004, 0.015)
  spike_noise_rate   0.002
  spike_noise_amp    (0.02, 0.08)
  tail_decay         (300.0, 1200.0)
  jitter_frac        0.5
  labels             ('P0', 'P1', 'P2', 'P3', 'P4')
  label_at_apex      True
  apex_search        (6, 40)
  min_apex_prom      0.01
  max_redraws        200
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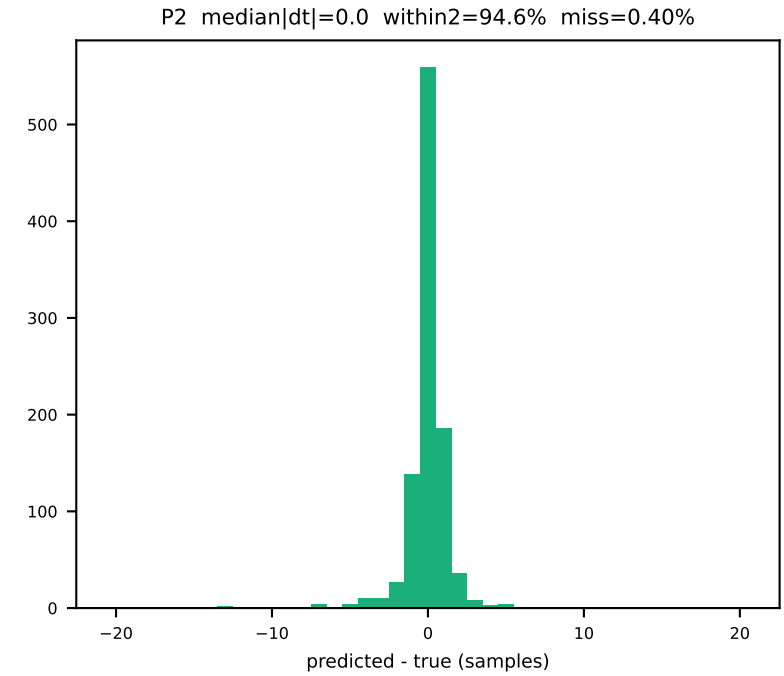

Toy waveforms with ground-truth peak positions
P0
P1
P2
P3
P4
V (a.u.)
sample
0
200
400
600
800
1000

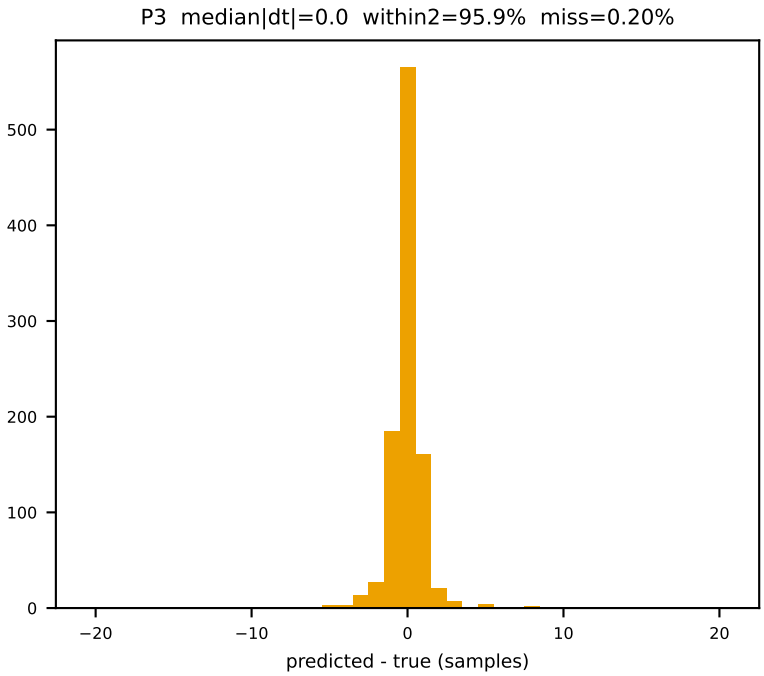

Validation traces: dashed = truth, solid = argmax of heatmap
dt = P0:+0 P1:+0 P2:+0 P3:+1 P4:+1 conf 0.97 0.97 0.98 0.96 0.93
P0
P1
P2
P3
P4
dt = P0:-2 P1:+0 P2:+0 P3:-1 P4:-1 conf 0.93 0.98 0.98 0.97 0.98
dt = P0:-1 P1:+0 P2:+0 P3:-2 P4:+0 conf 0.97 0.96 0.98 0.92 0.97
dt = P0:-2 P1:+0 P2:+2 P3:+0 P4:-1 conf 0.95 0.98 0.94 0.98 0.97
sample

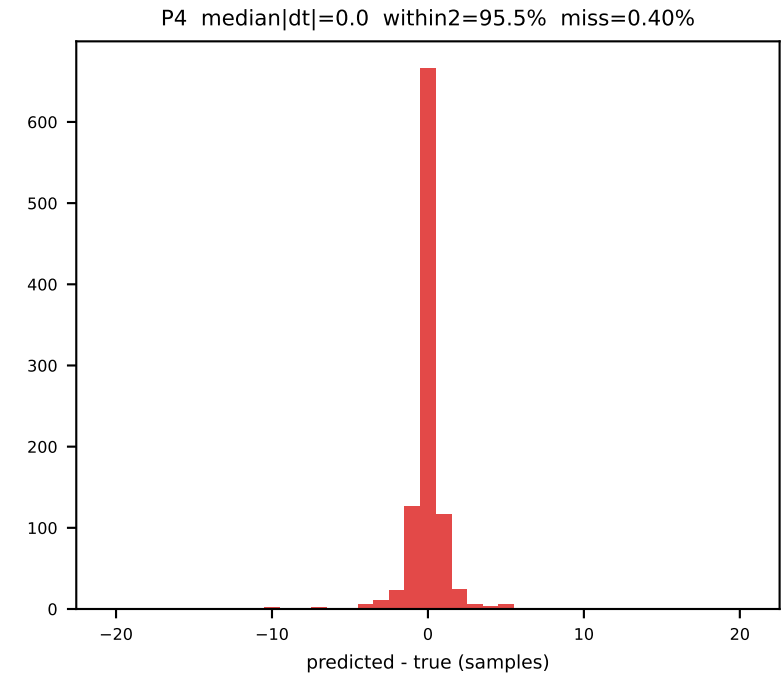
Position error on 1000 held-out toys

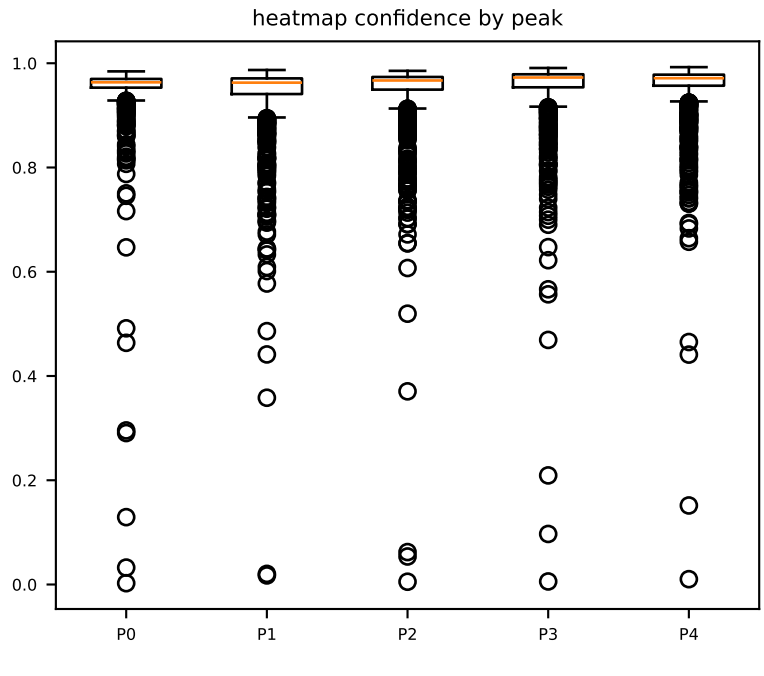


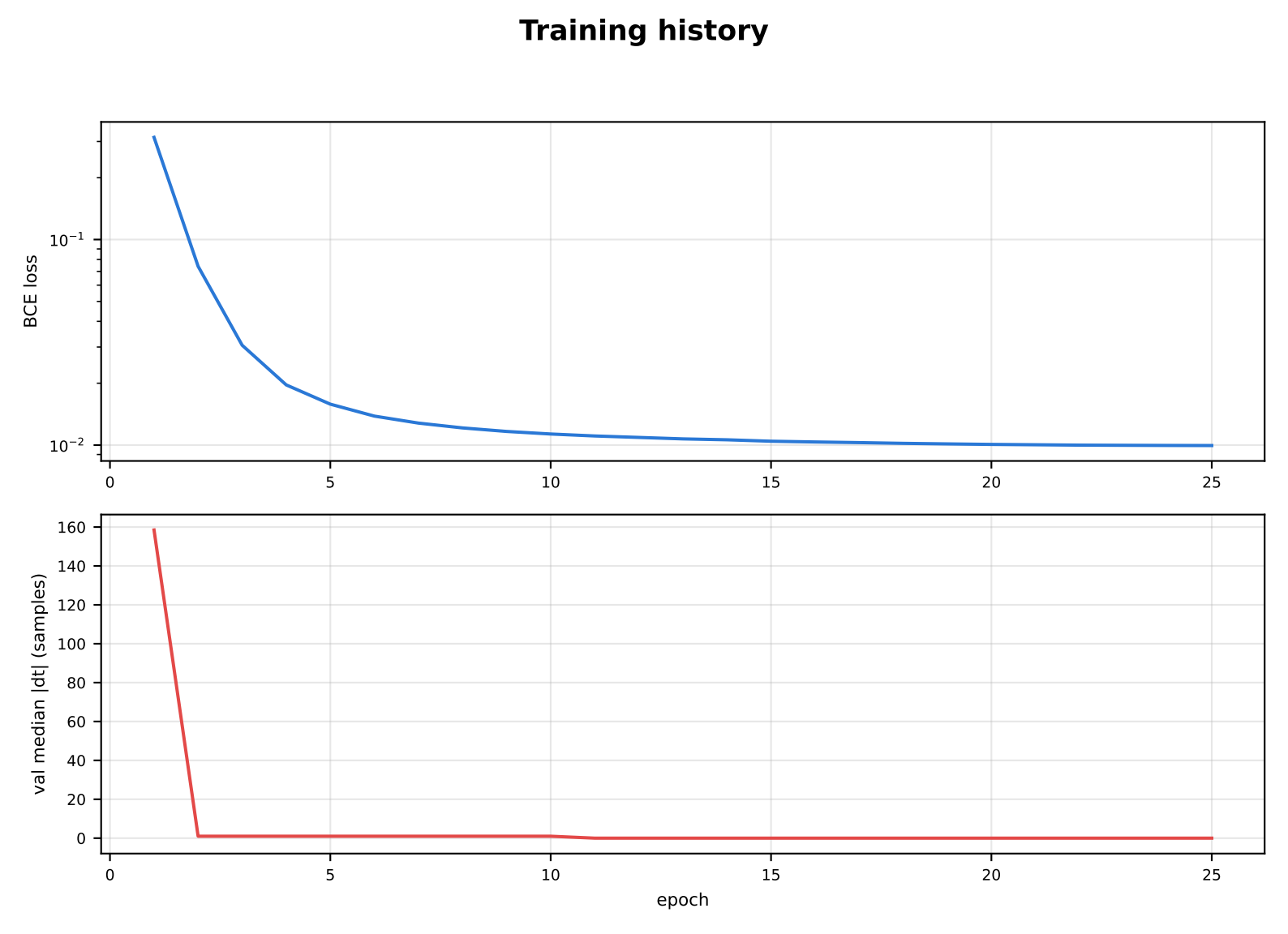




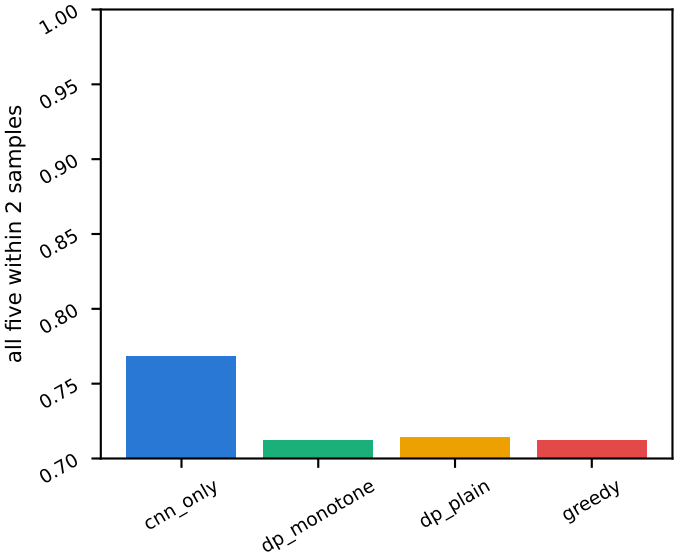


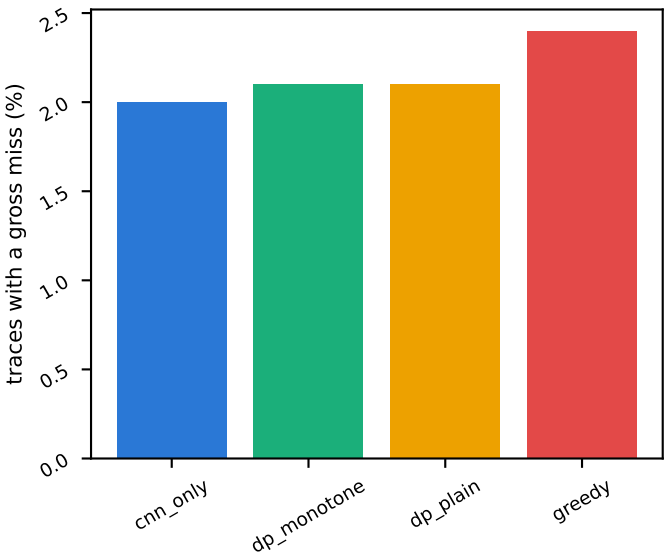


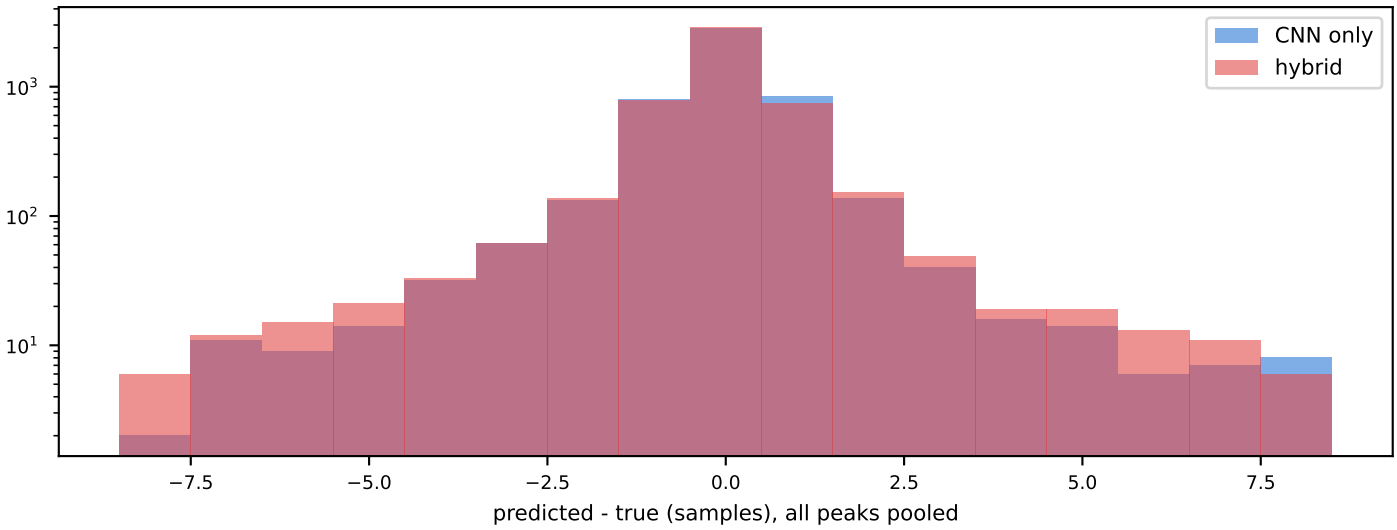



Training history
BCE loss
10^{-1}
10^{-2}
0
5
10
15
20
25
val median |dt| (samples)
160
140
120
100
80
60
40
20
0
epoch

Stage 2: candidate scan + classifier








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WHY THE HYBRID DOES NOT WIN ON TOYS

  CNN answer inside the candidate set : 1000/1000 traces
  CNN chain violating monotone apex   : 1/1000 traces
  DP fallbacks (no valid chain)       : 0/1000
  mean candidates per trace           : 181.7

  The scan's local maxima are systematically displaced from the pulse centre:
  the rising floor step and the previous peak's decay pull the apex off. Scan-only
  candidate recall (true peak within 3 samples) therefore caps the method:

    prominence   candidates/trace   recall within 3 samples
         0.004              253.8                      0.999
         0.010              180.8                      0.995
         0.020               98.7                      0.971
         0.040               32.5                      0.909
         0.080                8.0                      0.781

  Adding the CNN's own heatmap maxima to the candidate set lifts recall to ~0.997,
  and the CNN answer is then always reachable - but the stage-2 classifier still
  ranks worse than the heatmap it is fed, because that prior is one feature among 22
  spread over ~170 candidates. Decoder choice is not the cause: monotone DP, plain DP
  and greedy per-channel all land within 0.005 of each other.

RECOMMENDATION
  Keep the CNN-only path as the default. Keep the scan + classifier code as a switch to
  re-test on real traces, where it may matter if the true peak is DEFINED as the sample
  maximum, or if the CNN mislocalises on shapes the toys do not cover.
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